

Sahaya Small steps. Meaningful learning.

An English–Kannada browser learning prototype
for educator-guided foundational practice.

PO5 / PO6 / SDG 4

DAKSHINA KANNADA

20–25 minute lesson • demonstration • discussion

AUTHOR

Dr. Lokesh M R

Professor

Department of Information
Science & Engineering

**A J Institute of Engineering
and Technology**

Mangaluru, Karnataka, India

Affiliated to VTU, Belagavi

Dr. Lokesh M R

Professor • Department of Information Science & Engineering

A J Institute of Engineering and Technology
Mangaluru, Karnataka, India

Affiliated to Visvesvaraya Technological University (VTU), Belagavi

22+

Years of experience

Ph.D.

Computer Science &
Engineering

Sathyabama Institute of
Science and Technology

M.Tech

Information
Technology

NIE, Mysore

B.E.

Electronics & Communication
Engineering

PES College of Engineering,
Mandya

One course mandate. A clear boundary.

Sl. No.	Broad Areas/Project titles	Measurable engineering outcome	Specific POs	SDGs
1	Computer-Based Learning Tool for Mentally Challenged Students	Design and develop a web/browserbased learning tool as per the requirements of the special school.	5, 6	4

The quoted syllabus row is preserved exactly. New project language is learner-centred and respectful.

PO5 · tools and limits | PO6 · responsible community use | SDG 4 · inclusive education

School-specific requirements have not yet been collected or approved.

Start with the learner—not with the model.

Three questions guide the baseline design; a school must validate the answers.

Can I follow it?

- Short prompts
- Familiar visual examples
- English + Kannada

Can I control it?

- No response timer
- Supportive retries
- Small practice sessions

Can I use it safely?

- No learner account
- Local practice records
- Clear exit and reset

Boundary: no diagnosis, ability ranking, cloud learner database, or claim of improved learning.

Local context: Dakshina Kannada

Karnataka, India • district headquarters: coastal Mangaluru



2021 BOUNDARY VINTAGE

District context.
No school directory.

Mangaluru point:
approximate

School locations: not
collected

Boundary licence: ODbL 1.0

30 cards. Two kinds of content.

Everyday practice is authored; the optional handwriting module uses real secondary data.

5

Numbers

5

Colours

5

Shapes

5

**Everyday
life**

10

**Handwritten
digits**

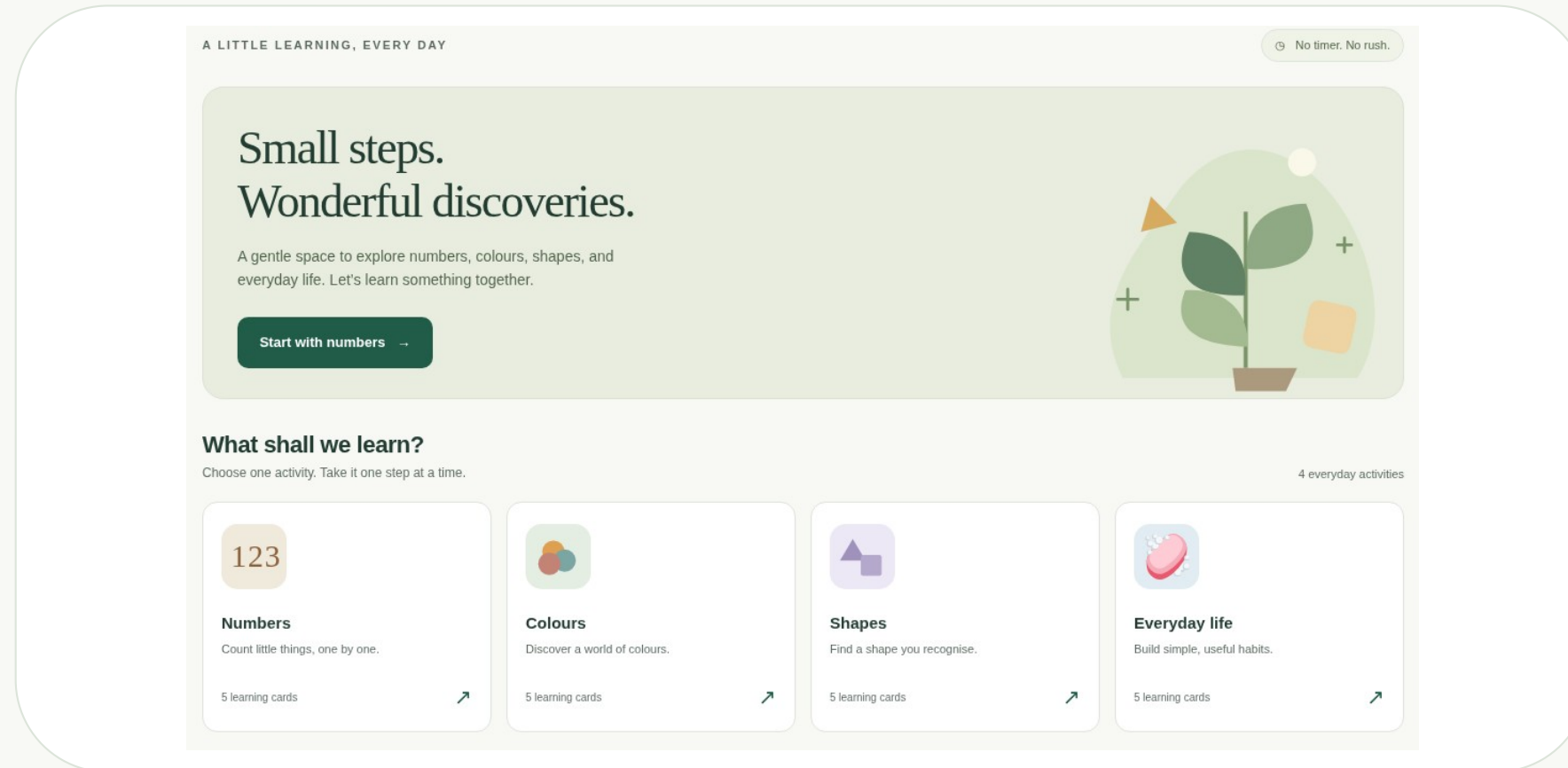
20 author-written everyday cards

10 UCI cards

All cards have English and Kannada prompts. Kannada and teaching suitability require educator review.

A calm starting point

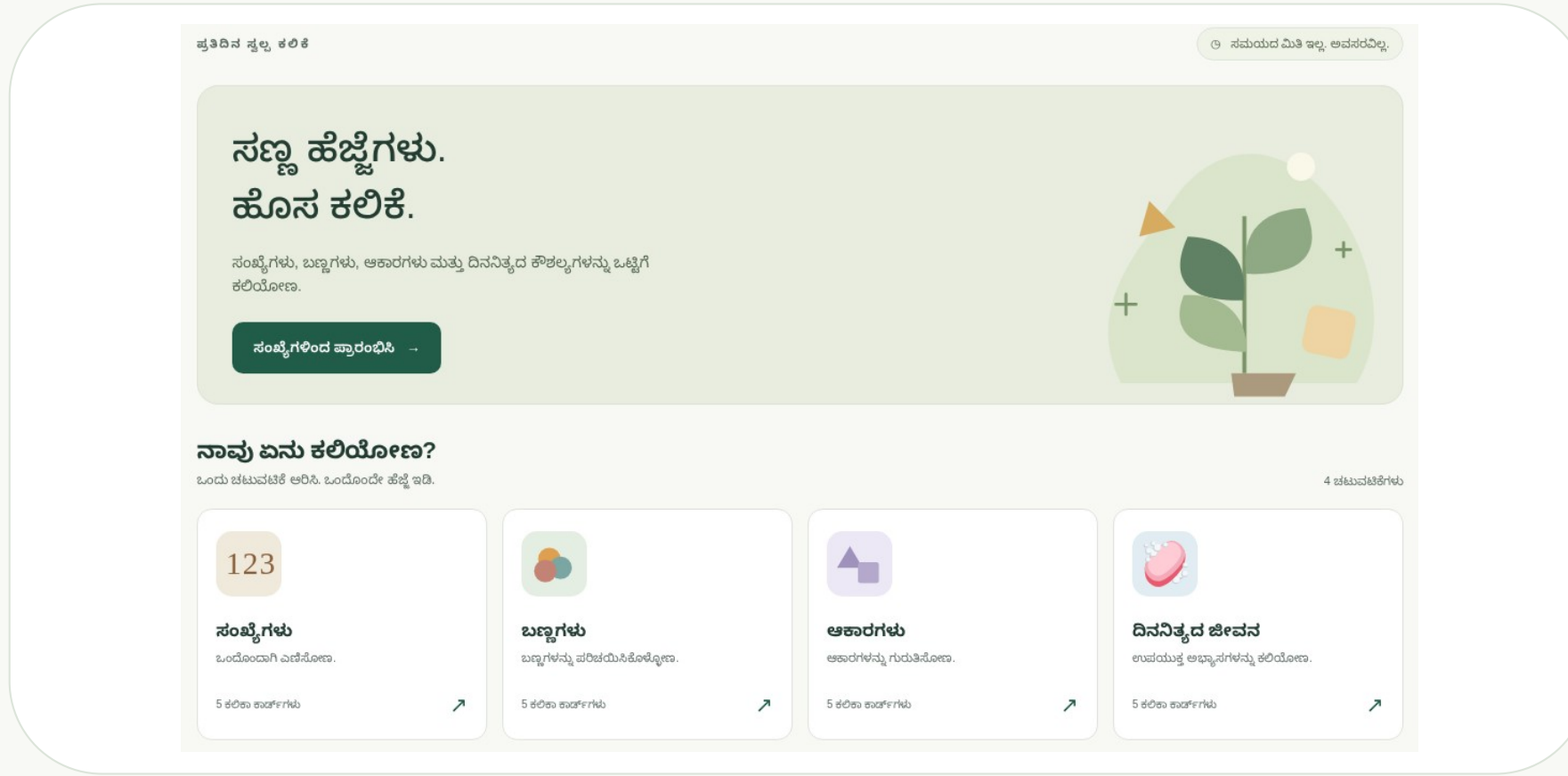
Actual application screen from the delivered project—not a classroom photograph.



- 1 Choose**
Four everyday activities.
- 2 Try**
No time pressure.
- 3 Continue**
Short, repeatable practice.

Same activity. A choice of language.

Learner-facing interface and prompts support English and Kannada.



English ⇌
Kannada

Bundled font

No font CDN needed

Review still needed:

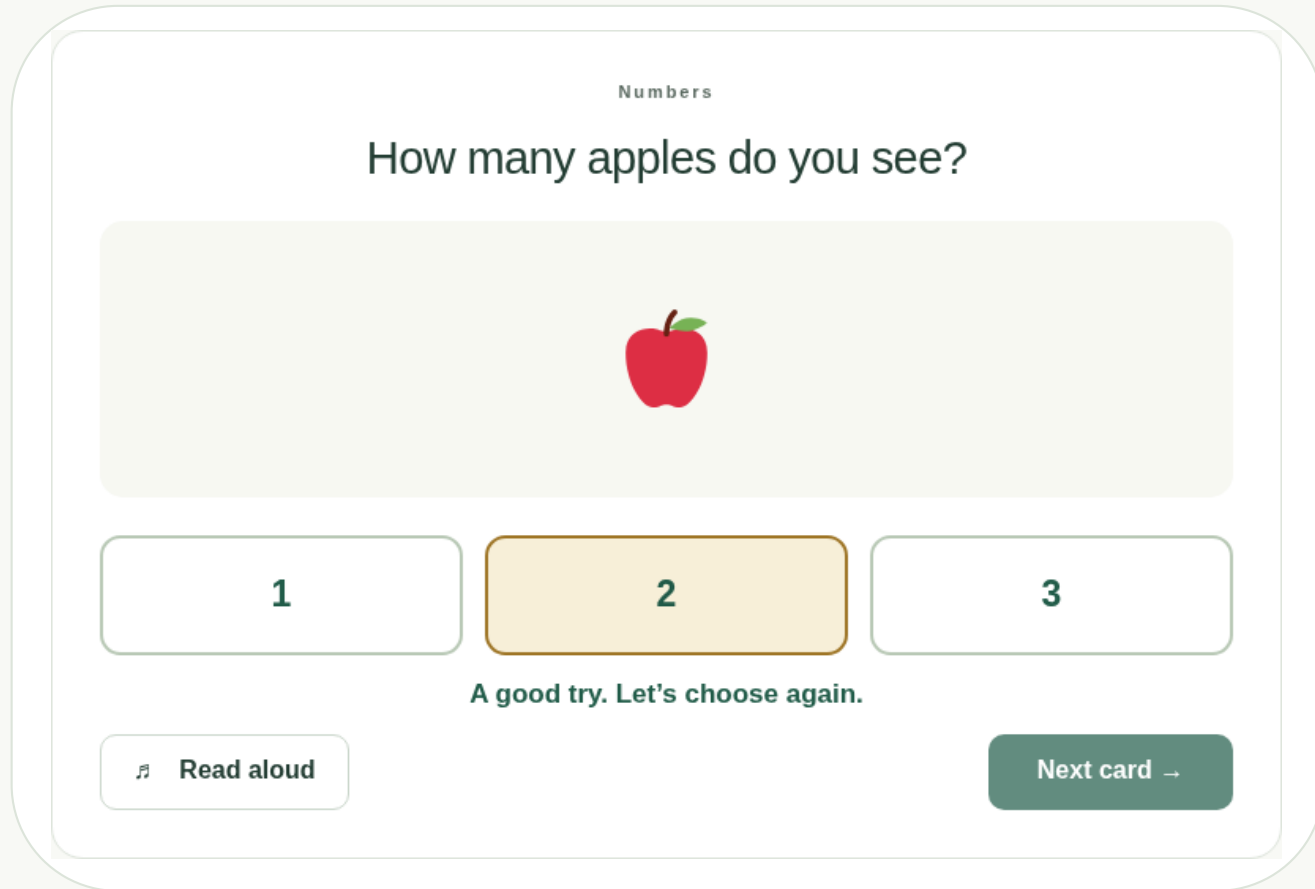
Kannada wording

Local examples

Individual suitability

Practice supports another try.

The model does not decide whether a learner can continue.



- 1 Choose**
Select an activity.
- 2 Respond**
Pick an answer; no timer.
- 3 Receive feedback**
Retry, or confirm success.
- 4 Continue**
Next card or a break.

3 or 5 cards • deterministic rotation, not adaptive AI

Accessible design is a starting point.

Technical checks do not establish universal accessibility or individual suitability.

Adjust the presentation

Larger text, higher contrast, visible keyboard focus.

Reduce pressure

No timed answers; short sessions; supportive retries.

Offer alternatives

Colour, shape and digit recognition are visual tasks.

LEARNING PREFERENCES

Make yourself comfortable

Larger text ☒

Higher contrast ☒

Remember progress on this device ☐

Off by default. No names, accounts, or cloud records. Shared devices combine everyone's practice.

Cards in one practice 5

Done

CLASS PROMPT Which activities would need an alternative for a learner who cannot see the screen?

Practice stays local by design.

No names, diagnoses, learner accounts, analytics, or learner uploads.



Shared-device records are not per-learner records. Clear them before changing users.

Browser voices and external links may use the network; hosting still sees ordinary request metadata.

Use open resources—with provenance.

A licence and a source are part of the engineering result.

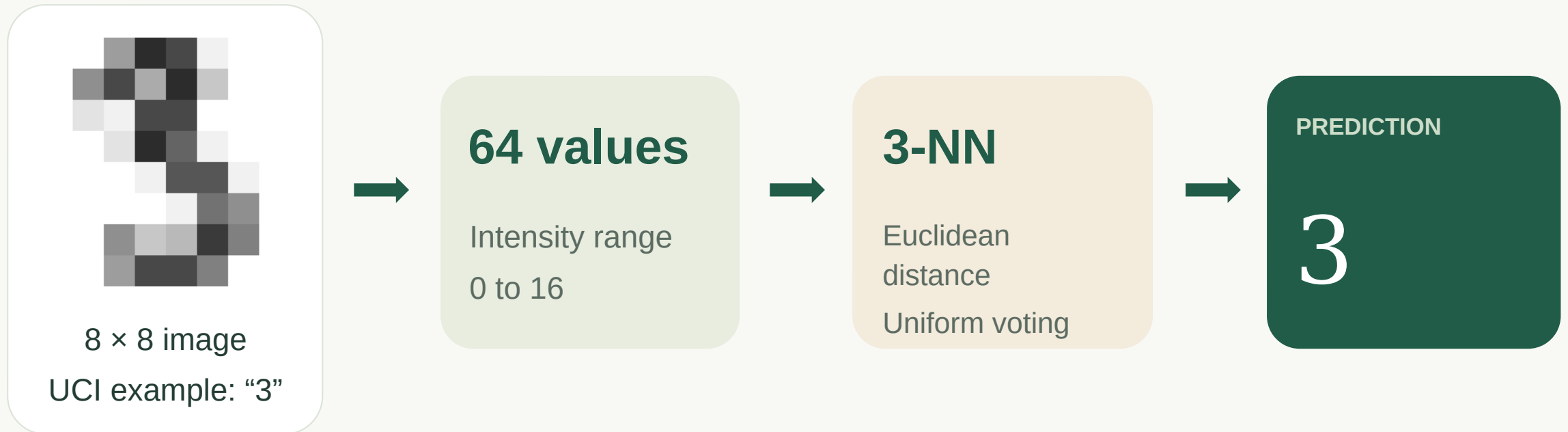
Resource	How this project uses it	Licence
UCI handwritten digits	Model fit/test; 10 optional digit cards	CC BY 4.0
geoBoundaries IND ADM2	Extracted district geometry and map	ODbL 1.0
Twemoji graphics	Apple, banana, soap and water SVGs	CC BY 4.0
scikit-learn 1.6.1	Executed 3-NN reference implementation	BSD-3-Clause
Noto Sans Kannada	Bundled Kannada font	SIL OFL 1.1

IndicTrans2: documented as an optional model—not downloaded, executed, or integrated.

Finders: UCI • Google Dataset Search • Hugging Face • OGD India • Kaggle

A transparent model: three nearest neighbours

Classifies repository digit images—not learners, disability, or learning progress.



Python evaluates the full source split. A small JavaScript equivalent runs examples locally in the educator view.

97.83% measures digit recognition.

It does not measure learner improvement.

97.83%

1,758 / 1,797 test digits classified correctly



39 errors • original fixed split • no test-label tuning

3,823 training rows

Original UCI training file.

1,797 test rows

Original held-out test file.

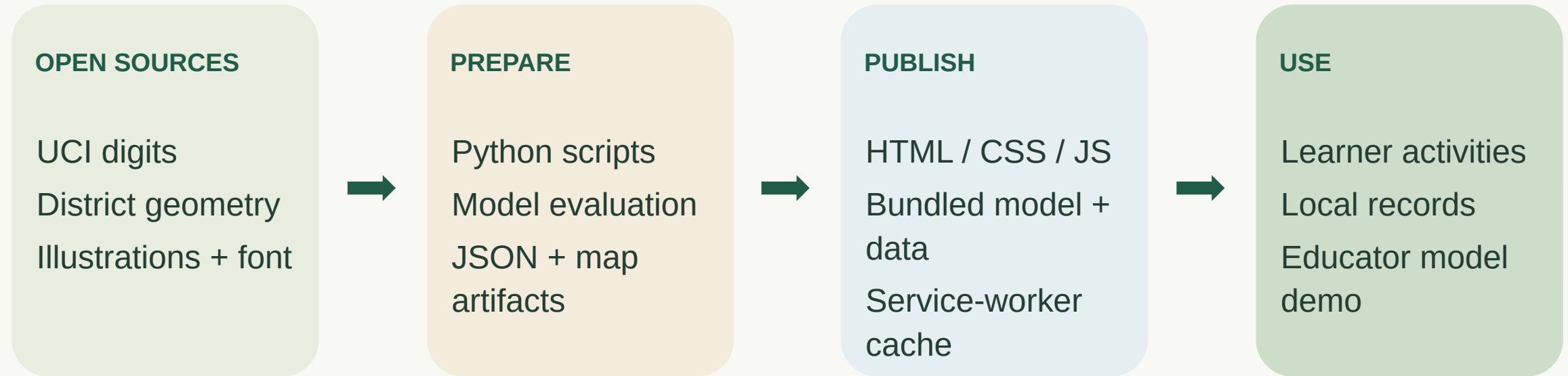
20 browser examples

Prediction parity checks, not a full browser evaluation.

Do not report this percentage as a learning gain, disability score, or Kannada handwriting accuracy.

A small static architecture

Build-time data processing is separate from learner practice.



No application backend, database, or mandatory API key.

Offline only after the first successful cache.
Some voices and source links need internet.

Evidence, not just a working screenshot.

Recorded engineering checks from the delivered package.

9

Engine tests

Content, grading, CSV,
model examples.

19

Browser checks

Language, storage,
reset, mobile, offline.

15

Week programs

Two runs each; byte-
identical stdout.

22

Artifact hashes

Checked before and
after regeneration.

0 detected axe violations across 4 main views ≠ full accessibility certification.

Classroom demonstration: three minutes

Use the live app; slides 7–10 are the screenshot fallback.

- 1 Open and orient**
Home → Numbers. Point out the absence of a timer.
- 2 Try, retry, continue**
Choose one wrong answer, then the correct one.
- 3 Switch and adjust**
Toggle Kannada; show larger text and contrast.
- 4 Inspect the evidence**
Progress → CSV/reset. Educators → model. Our district → map.

Before class: open the app once while online, verify assets, and keep a local copy available.

Git → Render: prepared, not published.

The package includes local history, a static-site Blueprint, and deployment instructions.

1

Review locally

Run and test the delivered source.

2

Push your Git repo

Use your own account; keep secrets out.

3

Connect Render

New → Blueprint; review render.yaml.

4

Verify HTTPS

Test Kannada, CSV, model, map and offline.

Build: node scripts/build.mjs | Publish: public | Secrets: none

Remote Git push and Render account deployment have not been performed.

The 15-week course journey

A proposed schedule—not an attendance or fieldwork record.

WEEKS 1–5

Frame the problem

Ethics and scope

Community questions

Requirements and sources

WEEKS 6–10

Design and build

Plan and feasibility

Data, model and map

Browser prototype

WEEKS 11–15

Validate and hand over

Technical tests

Approved fieldwork, if ready

Report, viva and deployment

Planned effort: 30 hours term work + 30 hours self-learning = 60 hours

A delivered output is not a measured outcome.

DELIVERED ENGINEERING

Working bilingual app
Sourced map and datasets
Model and technical evidence
Git / Render handover files

BEFORE CLASSROOM ADOPTION

School-approved requirements
Kannada and access review
Permission, consent and assent
Supervised trial and feedback

Learner improvement: not measured. School surveys and learner trials: none claimed.

Research, teaching & academic service

Over 22 years in teaching, research and administration across premier institutions in Karnataka.

RESEARCH SPECIALISATIONS

Cyber-Physical
Systems

Artificial Immune
Systems

Fault-Tolerant
Computing

Multi-Agent
Systems

LIFE MEMBER

CSI and ISTE

Publications & patent

Published extensively in Scopus-indexed and UGC-approved journals; holds a granted patent.

Supervision & funded research

Supervises Ph.D. scholars under VTU; contributed to KSCST- and AICTE-funded research projects.

Invited talks

FDPs, workshops and conferences.

Contact & scholarly profiles

Dr. Lokesh M R • select a link to open the corresponding profile.

EMAIL

lokeshmrmysore@gmail.com

MOBILE

[+91 9964050926](tel:+919964050926)

WEB OF SCIENCE ID

JCF-0012-2023

Professor, ISE • AJIET, Mangaluru

ORCID

[0000-0003-0025-7941](https://orcid.org/0000-0003-0025-7941)

SCOPUS

[Author ID: 57687382800](https://scopus.com/authid/detail/authid/57687382800)

GOOGLE SCHOLAR

[CCfMxaMAAAAJ](https://scholar.google.com/citations?user=CCfMxaMAAAAJ)

LINKEDIN

[lokesh-m-r-b6880b169](https://www.linkedin.com/in/lokesh-m-r-b6880b169)

GITHUB

[lokeshmr23](https://github.com/lokeshmr23)

VIDWAN

[Profile: 344450](https://www.vidwan.in/profile/344450)

Acknowledgement & dedication

The author expresses gratitude to:

VTU, Belagavi

For the curriculum framework.

A J Institute of Engineering and Technology

For infrastructure.

Deep Learning textbook

Ian Goodfellow, Yoshua Bengio and Aaron Courville.

All students who provided feedback

For their feedback.

Family

For constant support and encouragement.

DEDICATION

To all VTU
students who
aspire to excel in
their
examinations.

What would you validate before the first class?

The learner

Language, input, sensory needs.

The evidence

Source quality, tests, model limits.

The setting

Permission, device, support, privacy.

Build transparently. Validate locally. Report only what the evidence supports.

Report Chs. 13–15 and Appendix C; no new impact or deployment claim

References and reproducibility

Source files are included in the project package; repository links below are clickable.

PRIMARY DELIVERABLES	Community-Project-Sahaya-Continuation.pdf • 48 pages
UCI HANDWRITTEN DIGITS	Sahaya-Project-Package.zip • code, data, evidence, Git bundle Alpaydin & Kaynak, 1998 • DOI: 10.24432/C50P49 • CC BY 4.0
DISTRICT GEOMETRY + CONTEXT	geoBoundaries IND ADM2, 2021 • ODbL 1.0; official district profile
OPEN-SOURCE MODEL	scikit-learn 1.6.1 • KNeighborsClassifier • BSD-3-Clause
OPTIONAL MODEL ONLY	AI4Bharat IndicTrans2 En–Indic 200M • not integrated
DEPLOYMENT	Render Blueprint documentation • account setup remains pending

Asset credits: Twemoji / Twitter contributors (CC BY 4.0); Noto Sans Kannada (SIL OFL 1.1).